

## Module 4

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| <b>Module-4</b> | Project Financing, Capital structure, Sources of finance, internal accrual, securities, term loans, working capital, Equity and Debt, Venture capital and private equity. |
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## One-Mark Questions with Answers

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### Question 1

**Define the term 'Cost of Capital' in one sentence.**

**Answer:**

The cost of capital is the minimum rate of return that a company must earn on its investments to maintain its market value and satisfy the expectations of its investors.

### Question 2

**What does the acronym WACC stand for?**

**Answer:**

Weighted Average Cost of Capital

### Question 3

**State the formula for calculating the after-tax cost of debt when irredeemable debentures are issued.**

**Answer:**

$$K_d = \{I \times (1 - T)\} / NP$$

Where I = Interest, NP = Net Proceeds, T = Tax Rate

### Question 4

**According to the Gordon's Dividend Discount Model, what is the formula for the cost of equity (Ke)?**

**Answer:**

$$K_e = \{D_1/P_0\} + g$$

Where  $D_1$  = Expected dividend next year,  $P_0$  = Current market price,  $g$  = Constant growth rate

## **Question 5**

**What is the formula for the Capital Asset Pricing Model (CAPM) to calculate the cost of equity?**

**Answer:**

$$K_e = R_f + \beta(R_m - R_f)$$

Where  $R_f$  = Risk-free rate,  $\beta$  = Beta,  $R_m$  = Market return

## **Question 6**

**Why is the cost of debt calculated on an after-tax basis?**

**Answer:**

Interest on debt is a tax-deductible expense, which reduces the taxable income of the company, creating a tax shield that lowers the effective cost of debt.

## **Question 7**

**Which component of capital does not have an explicit cost: Equity shares, Preference shares, or Retained earnings?**

**Answer:**

Retained earnings (They have an implicit opportunity cost but no explicit cost like dividend or interest)

## **Question 8**

**What are flotation costs?**

**Answer:**

Flotation costs are the expenses incurred by a company when issuing new securities, such as underwriting fees, legal fees, printing costs, and registration fees.

## **Question 9**

**State whether the following statement is True or False:**

**"The cost of preference shares is tax-deductible."**

**Answer:**

False (Preference dividends are paid from after-tax profits and are not tax-deductible)

## **Question 10**

**If the risk-free rate is 5%, the market return is 12%, and beta is 1.5, what is the cost of equity as per CAPM?**

**Answer:**

$$K_e = 5\% + 1.5 \times (12\% - 5\%) = 5\% + 1.5 \times 7\% = 5\% + 10.5\% = 15.5\%$$

## **Question 11**

**What is the significance of the Weighted Average Cost of Capital (WACC) in capital budgeting?**

**Answer:**

WACC serves as the hurdle rate or minimum required rate of return for evaluating investment projects; a project is accepted only if its return exceeds the WACC.

## **Question 12**

**Between book value weights and market value weights, which is theoretically superior for calculating WACC?**

**Answer:**

Market value weights are theoretically superior as they reflect the current economic value of the firm's capital structure.

### Question 13

**Define 'Marginal Cost of Capital' (MCC).**

**Answer:**

Marginal Cost of Capital is the cost of raising an additional rupee of new capital; it represents the weighted average cost of the next unit of financing.

### Question 14

**If a company's cost of equity is 16% and the cost of debt (after tax) is 8%, with a debt-to-equity ratio of 1:1, what is the approximate WACC? (Assume no preference shares)**

**Answer:**

Weights: Debt = 0.5, Equity = 0.5

$$WACC = (0.5 \times 16\%) + (0.5 \times 8\%) = 8\% + 4\% = 12\%$$

### Question 18

**A company issues 10% debentures of ₹100 each at par. The tax rate is 30%. Calculate the after-tax cost of debt.**

**Answer:**

$$K_d = 10\% \times (1 - 0.30) = 10\% \times 0.70 = 7\%$$

### Question 19

**What is the impact of an increase in the corporate tax rate on the after-tax cost of debt?**

**Answer:**

The after-tax cost of debt decreases because the tax shield (interest  $\times$  tax rate) increases with higher tax rates.

### Question 20

**Define 'Optimal Capital Structure' in relation to the cost of capital.**

**Answer:**

Optimal capital structure is the mix of debt and equity that minimizes the overall Weighted Average Cost of Capital (WACC) and maximizes the value of the firm.

## Three-Mark Questions with Answers

### Question 1

**What is Weighted Average Cost of Capital (WACC)? Why is it considered the minimum required rate of return for a company's investment projects?**

**Answer:**

**Definition:**

Weighted Average Cost of Capital (WACC) is the average rate of return a company is expected to pay to all its security holders (equity shareholders, preference shareholders, and debt holders), weighted by the proportion of each source of capital in the company's capital structure.

**Why WACC is the Minimum Required Rate of Return:**

- **Cost of Financing:** WACC represents the blended cost of all funds used to finance operations and investments. Any project must earn at least this rate to cover the cost of raising capital.
- **Shareholder Wealth Maximization:** If a project earns less than WACC, it fails to provide the required return to investors, leading to a decline in the firm's market value.
- **Opportunity Cost:** WACC reflects the opportunity cost of capital for investors. Therefore, it serves as the minimum acceptable return (hurdle rate) for new investments.

### Question 2

**Distinguish between the Cost of Debt and the Cost of Equity. Why is the cost of debt calculated on an after-tax basis while the cost of equity is not?**

**Answer:**

|                      |                                               |                                           |
|----------------------|-----------------------------------------------|-------------------------------------------|
| Basis                | Cost of Debt                                  | Cost of Equity                            |
| <b>Return</b>        | Fixed interest payment                        | Variable dividend or capital appreciation |
| <b>Risk</b>          | Lower risk (secured, priority in liquidation) | Higher risk (residual claim)              |
| <b>Tax Treatment</b> | Tax-deductible expense                        | Not tax-deductible                        |
| <b>Cost</b>          | Generally lower                               | Generally higher                          |

#### **Reason for After-tax Cost of Debt:**

Interest paid on debt is a legitimate business expense deductible from taxable income. This creates a tax shield that reduces the effective cost. For example, if interest rate is 10% and tax rate is 30%, the after-tax cost is only 7%. This tax benefit is not available for equity dividends, which are paid from after-tax profits. Hence, cost of equity is calculated on a pre-tax basis.

### **Question 3**

**A company's equity share is currently traded at ₹120 per share. The company is expected to pay a dividend of ₹6 per share next year, and dividends are expected to grow at a constant rate of 5% per annum. Calculate the cost of equity using the Gordon's Dividend Discount Model.**

**Answer:**

**Given:**

- Current Market Price ( $P_0$ ) = ₹120
- Expected Dividend ( $D_1$ ) = ₹6
- Growth Rate ( $g$ ) = 5% = 0.05

**Formula:**

$$K_e = \{D_1/P_0\} + g$$

**Calculation:**

$$K_e = (6/120) + 0.05 = 0.05 + 0.05 = 0.10 \text{ or } 10\%$$

**Result:** The cost of equity is **10%**.

## Question 4

**The risk-free rate of return is 6%, the expected market return is 14%, and a company's beta ( $\beta$ ) is 1.3. Calculate the cost of equity using the Capital Asset Pricing Model (CAPM). Briefly explain what beta represents in this context.**

**Answer:**

**Given:**

- Risk-free Rate ( $R_f$ ) = 6%
- Market Return ( $R_m$ ) = 14%
- Beta ( $\beta$ ) = 1.3

**Formula:**

$$K_e = R_f + \beta(R_m - R_f)$$

**Calculation:**

$$K_e = 6\% + 1.3 \times (14\% - 6\%) = 6\% + 1.3 \times 8\% = 6\% + 10.4\% = 16.4\%$$

**Result:** The cost of equity is **16.4%**.

**What Beta Represents:**

Beta measures the systematic risk or volatility of a stock relative to the overall market. A beta of 1.3 indicates that the stock is 30% more volatile than the market. If the market moves up by 1%, the stock is expected to move up by 1.3%, and vice versa.

## Question 5

A company has issued 9% irredeemable debentures of ₹100 each. The debentures are currently traded at ₹90 in the market. The corporate tax rate is 35%. Calculate the after-tax cost of debt.

**Answer:**

**Given:**

- Interest Rate = 9% → Interest (I) = ₹9 per debenture
- Net Proceeds (NP) = ₹90
- Tax Rate (T) = 35% = 0.35

**Formula:**

$$K_d = \{I/NP\} \times (1 - T)$$

**Calculation:**

$$K_d = (9/90) \times (1 - 0.35) = 0.10 \times 0.65 = 0.065 \text{ or } 6.5\%$$

**Result:** The after-tax cost of debt is **6.5%**.

## Question 6

A company issues 10% preference shares of ₹100 each at a discount of 5%. The shares are redeemable after 8 years at par. Calculate the cost of preference shares using the approximation method. (Ignore flotation costs)

**Answer:**

**Given:**

- Face Value (FV) = ₹100
- Redemption Value (RV) = ₹100
- Dividend (D) = 10% of ₹100 = ₹10
- Net Proceeds (NP) = ₹100 – 5% discount = ₹95
- Life (n) = 8 years

**Approximation Formula:**

$$K_p = [D + (RV - NP)/n] / (RV + NP)/2$$



**Calculation:**

$$K_p = [10 + (100 - 95)/8] / (100 + 95)/2 = [10 + (5/8)] / (195/2) = (10 + 0.625) / 97.5 = 10.625 / 97.5 = 0.10897 \text{ or } 10.90\%$$

**Result:** The cost of preference shares is **10.90%**.

**Question 7**

**Explain the concept of flotation costs. How does the cost of newly issued equity differ from the cost of retained earnings?**

**Answer:**

**Flotation Costs:**

Flotation costs are the expenses incurred by a company when issuing new securities. These include underwriting fees, legal fees, registration fees, printing costs, and other administrative expenses. These costs reduce the net proceeds received by the company from the issue.

**Difference between Cost of New Equity and Cost of Retained Earnings:**

| Aspect                 | Cost of New Equity                       | Cost of Retained Earnings                      |
|------------------------|------------------------------------------|------------------------------------------------|
| <b>Flotation Costs</b> | Includes flotation costs                 | No flotation costs                             |
| <b>Net Proceeds</b>    | Lower (Issue Price minus Flotation Cost) | Equal to current market price                  |
| <b>Cost</b>            | Higher                                   | Lower (by the amount of flotation cost effect) |
| <b>Formula</b>         | $K_e = D_1 / (P_0 - F) + g$              | $K_r = D_1 / P_0 + g$                          |

Retained earnings represent funds already belonging to shareholders and involve no issuance expenses, making them a cheaper source of equity financing compared to new equity issues.

## Question 8

**Differentiate between book value weights and market value weights used in calculating WACC. Which approach is theoretically superior and why?**

**Answer:**

| Basis            | Book Value Weights                    | Market Value Weights              |
|------------------|---------------------------------------|-----------------------------------|
| <b>Basis</b>     | Based on historical accounting values | Based on current market prices    |
| <b>Reflects</b>  | Past financing decisions              | Current economic conditions       |
| <b>Stability</b> | More stable over time                 | Fluctuates with market conditions |
| <b>Relevance</b> | May not reflect true opportunity cost | Reflects true opportunity cost    |

**Theoretically Superior Approach:**

**Market Value Weights** are theoretically superior because:

- They represent the true economic value of the firm's capital structure.
- Investors evaluate returns based on current market prices, not historical costs.
- WACC calculated with market weights better reflects the current opportunity cost of capital for investment decisions.

## Seven-Mark Questions with Answers

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### Question 1

**What do you understand by the 'Cost of Capital'? Explain the significance of the Weighted Average Cost of Capital (WACC) in capital budgeting decisions.**

**Answer:**

**Cost of Capital** refers to the minimum rate of return that a firm must earn on its investments to maintain its market value and satisfy the expectations of its investors

(equity shareholders, preference shareholders, and debt holders). It represents the overall cost of financing a firm's operations and investments.

### Significance of WACC in Capital Budgeting:

- **Hurdle Rate:** WACC serves as the discount rate or hurdle rate for evaluating investment proposals. A project is accepted only if its Internal Rate of Return (IRR) exceeds the WACC.
  - **Capital Budgeting Decisions:** It helps in selecting the optimal mix of projects that maximize shareholder wealth.
  - **Performance Evaluation:** WACC is used to evaluate whether the firm is generating returns above its cost of financing.
  - **Capital Structure Decisions:** It aids in determining the optimal capital structure that minimizes the overall cost of capital.
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## Question 2

**A company has the following capital structure:**

- Equity Share Capital (1,00,000 shares of ₹10 each): ₹10,00,000
- Retained Earnings: ₹5,00,000
- 10% Debentures (₹100 each): ₹5,00,000
- 12% Preference Shares (₹100 each): ₹2,00,000

**The market price of an equity share is ₹25. The company is expected to pay a dividend of ₹2 per share next year, which is expected to grow at 6% per annum. The tax rate is 40%. Calculate the WACC based on book value weights and market value weights. (4 Marks)**

**Answer:**

### Step 1: Calculate Cost of Each Component

- **Cost of Equity (Ke):** Using Gordon's Model

$$K_e = (D_1/P_0) + g = (2/25) + 0.06 = 0.08 + 0.06 = 0.14 \text{ or } 14\%$$

- **Cost of Retained Earnings (Kr):**  $K_r = K_e = 14\%$  (opportunity cost)
- **Cost of Debentures (Kd):** After-tax

$$K_d = \text{Interest Rate} \times (1 - T) = 10\% \times (1 - 0.40) = 6\%$$

- **Cost of Preference Shares (Kp):**

$$K_p = \text{Dividend} / \text{Market Price} = 12 / 100 = 12\%$$

(Note: Market price assumed equal to face value as not given separately)

### Step 2: WACC using Book Value Weights

| Source            | Book Value (₹)   | Weight        | Cost | Weighted Cost        |
|-------------------|------------------|---------------|------|----------------------|
| Equity            | 10,00,000        | 0.4545        | 14%  | 6.363%               |
| Retained Earnings | 5,00,000         | 0.2273        | 14%  | 3.182%               |
| Debentures        | 5,00,000         | 0.2273        | 6%   | 1.364%               |
| Preference        | 2,00,000         | 0.0909        | 12%  | 1.091%               |
| <b>Total</b>      | <b>22,00,000</b> | <b>1.0000</b> |      | <b>WACC = 12.00%</b> |

### Step 3: WACC using Market Value Weights

- **Market Value of Equity:** 1,00,000 shares × ₹25 = ₹25,00,000
- **Market Value of Retained Earnings:** Assumed same as book = ₹5,00,000 (no separate market)
- **Market Value of Debentures:** 5,000 debentures × ₹100 = ₹5,00,000
- **Market Value of Preference:** 2,000 shares × ₹100 = ₹2,00,000
- **Total Market Value:** ₹37,00,000

| Source            | Market Value (₹) | Weight | Cost | Weighted Cost |
|-------------------|------------------|--------|------|---------------|
| Equity            | 25,00,000        | 0.6757 | 14%  | 9.460%        |
| Retained Earnings | 5,00,000         | 0.1351 | 14%  | 1.891%        |
| Debentures        | 5,00,000         | 0.1351 | 6%   | 0.811%        |
| Preference        | 2,00,000         | 0.0541 | 12%  | 0.649%        |

| Source       | Market Value (₹) | Weight        | Cost | Weighted Cost        |
|--------------|------------------|---------------|------|----------------------|
| <b>Total</b> | <b>37,00,000</b> | <b>1.0000</b> |      | <b>WACC = 12.81%</b> |

### Question 3

A company's equity shares are currently traded at ₹180 per share. The company expects to pay a dividend of ₹9 per share next year, and the dividends are expected to grow at a constant rate of 8% per annum.

- Calculate the cost of equity using the Gordon's Dividend Discount Model.
- If the Risk-Free Rate of return is 7%, the market rate of return is 15%, and the company's beta ( $\beta$ ) is 1.2, calculate the cost of equity using the Capital Asset Pricing Model (CAPM).
- As a financial analyst, which of the two computed costs would you consider more reliable and why?

**Answer:**

- $K_e = (D_1/P_0) + g = (9/180) + 0.08 = 0.05 + 0.08 = 0.13$  or 13%
- $K_e = R_f + \beta(R_m - R_f) = 7\% + 1.2 \times (15\% - 7\%) = 7\% + 1.2 \times 8\% = 7\% + 9.6\% = 16.6\%$
- The **CAPM approach (16.6%)** is generally considered more reliable for the following reasons:

**Risk Consideration:** CAPM explicitly incorporates systematic risk (beta), which reflects the stock's sensitivity to market movements. Gordon's model assumes constant growth, which may not hold true in reality.

**Market-Based:** CAPM uses market-determined inputs (risk-free rate, market risk premium), making it more objective and forward-looking.

**Limitations of Gordon's Model:** The dividend growth model assumes a constant growth rate perpetually and requires that the growth rate is less

than the cost of equity. It also fails for companies that do not pay dividends.

However, in practice, many analysts use an average of both methods to arrive at a more balanced estimate of the cost of equity.

## Question 4

**A manufacturing firm has issued the following securities. Compute the cost of each component, assuming a corporate tax rate of 35%.**

- (a) **Debentures:** 1,000, 12% debentures of ₹100 each, redeemable after 5 years at a premium of 5%. The debentures are currently sold at ₹96 each. (Use approximation method)
- (b) **Preference Shares:** 500, 10% preference shares of ₹100 each, redeemable after 4 years at par. The current market price of the preference share is ₹92. (Use approximation method)

**Answer:**

(a)

**Given:**

- Face Value (FV) = ₹100
- Redemption Value (RV) = ₹100 + 5% premium = ₹105
- Interest (I) = 12% of ₹100 = ₹12 per debenture
- Net Proceeds (NP) = ₹96 (market price)
- Tax Rate (T) = 35%
- Life (n) = 5 years

### **Step 1: Pre-tax Cost of Debt (Approximation Formula)**

$$\begin{aligned} K_{dpre} &= [I + (RV - NP)/n] / (RV + NP)/2 = [12 + (105 - 96)/5] / [105 + 96]/2 \\ &= [12 + (9/5)] / [201/2] = (12 + 1.8) / 100.5 \\ &= 13.8 / 100.5 = 0.1373 \text{ or } 13.73\% \end{aligned}$$

### **Step 2: After-tax Cost of Debt**

$$K_d = K_{dpre} \times (1 - T) = 13.73\% \times (1 - 0.35) = 13.73\% \times 0.65 = 8.92\%$$

(b)

**Given:**

- Face Value (FV) = ₹100
- Redemption Value (RV) = ₹100
- Dividend (D) = 10% of ₹100 = ₹10 per share
- Net Proceeds (NP) = ₹92
- Life (n) = 4 years

**Approximation Formula:**

$$\begin{aligned} K_p &= [D + (RV - NP)/n] / (RV + NP) \times 2 \\ &= \{10 + (100 - 92)/4\} / (100 + 92) \times 2 \\ &= [10 + (8/4)] / (192/2) \\ &= (10 + 2) / 96 \\ &= 12/96 = 0.125 \text{ or } 12.5\% \end{aligned}$$

**Note:** Preference dividends are not tax-deductible, so no tax adjustment is required.

## Question 5

**The following is the extract of the balance sheet of XYZ Ltd. as on 31st March:**

| Liabilities                     | Amount (₹)       | Assets         | Amount (₹)       |
|---------------------------------|------------------|----------------|------------------|
| Equity Share Capital (₹10 each) | 15,00,000        | Fixed Assets   | 25,00,000        |
| 10% Preference Share Capital    | 5,00,000         | Current Assets | 15,00,000        |
| 12% Debentures                  | 10,00,000        |                |                  |
| Retained Earnings               | 5,00,000         |                |                  |
| Current Liabilities             | 5,00,000         |                |                  |
| <b>Total</b>                    | <b>40,00,000</b> | <b>Total</b>   | <b>40,00,000</b> |

### Additional Information:

- The current market price of an equity share is ₹24. The last dividend paid was ₹1.60 per share, and dividends are expected to grow at 5% p.a.
- The tax rate is 40%.
- Preference share and debenture market prices are ₹90 and ₹105 respectively.

**Calculate the Weighted Average Cost of Capital (WACC) using market value weights.**

**Answer:**

#### Step 1: Calculate Number of Securities

- **Equity Shares:** ₹15,00,000 ÷ ₹10 = 1,50,000 shares
  - **Preference Shares:** ₹5,00,000 ÷ ₹100 = 5,000 shares
  - **Debentures:** ₹10,00,000 ÷ ₹100 = 10,000 debentures
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#### Step 2: Calculate Market Values

- **Equity:** 1,50,000 shares × ₹24 = ₹36,00,000
  - **Retained Earnings:** Market value not separately available; use book value = ₹5,00,000
  - **Preference Shares:** 5,000 shares × ₹90 = ₹4,50,000
  - **Debentures:** 10,000 debentures × ₹105 = ₹10,50,000
  - **Total Market Value:** ₹36,00,000 + ₹5,00,000 + ₹4,50,000 + ₹10,50,000 = ₹56,00,000
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#### Step 3: Calculate Cost of Each Component

- **Cost of Equity (Ke):** Using Gordon's Model

$$D_1 = D_0 \times (1+g) = 1.60 \times 1.05 = ₹1.68$$

$$K_e = \frac{D_1}{P_0} + g$$

$$= \frac{1.68}{24} + 0.05$$



$$=0.07+0.05=0.12 \text{ or } 12\%$$

- **Cost of Retained Earnings (Kr):**  $K_r = K_e = 12\%$
- **Cost of Preference Shares (Kp):**

$$K_p = \text{Dividend} / \text{Market Price}$$

$$=10/90=0.1111 \text{ or } 11.11\%$$

- **Cost of Debentures (Kd):** After-tax

$$K_d = \text{Interest} / \text{Market Price} \times (1 - T)$$

$$=12/105 \times (1 - 0.40)$$

$$=0.1143 \times 0.60$$

$$=0.0686 \text{ or } 6.86\%$$

#### Step 4: Calculate WACC using Market Value Weights

| Source            | Market Value (₹) | Weight        | Cost   | Weighted Cost        |
|-------------------|------------------|---------------|--------|----------------------|
| Equity            | 36,00,000        | 0.6429        | 12.00% | 7.715%               |
| Retained Earnings | 5,00,000         | 0.0893        | 12.00% | 1.072%               |
| Preference Shares | 4,50,000         | 0.0804        | 11.11% | 0.893%               |
| Debentures        | 10,50,000        | 0.1875        | 6.86%  | 1.286%               |
| <b>Total</b>      | <b>56,00,000</b> | <b>1.0000</b> |        | <b>WACC = 10.97%</b> |

## Question 6

**A company is considering a project that requires an initial investment of ₹50 lakhs. The company's existing capital structure consists of:**

- **Equity:** ₹30 lakhs (Cost: 18%)
- **Debt:** ₹20 lakhs (Cost: 9% post-tax)

**The company plans to raise the required ₹50 lakhs for the new project in a manner that maintains the same debt-to-equity ratio (D/E) as the existing structure. The cost of new debt will be 10% (post-tax) and the cost of new equity will be 20%.**

- (a) Calculate the Weighted Average Cost of Capital (WACC) for the new project.**
- (b) If the project is expected to generate an Internal Rate of Return (IRR) of 15%, should the company accept the project? Justify your answer.**

**Answer:**

### **Step 1: Determine Target Capital Structure**

Existing D/E Ratio = ₹20 lakhs / ₹30 lakhs =  $2/3 = 0.667$

Target Structure:

- Debt =  $0.667 / (1 + 0.667) = 0.667 / 1.667 = 0.40$  or 40%
- Equity =  $1 / (1 + 0.667) = 1 / 1.667 = 0.60$  or 60%

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### **Step 2: Calculate WACC for New Project**

Since the project will be financed with new capital maintaining the same structure:

$$\begin{aligned} \text{WACC} &= (W_d \times K_d) + (W_e \times K_e) \\ &= (0.40 \times 10\%) + (0.60 \times 20\%) \\ &= 4\% + 12\% = 16\% \end{aligned}$$

**(b)**

The project should be **rejected**.

**Justification:**

- The project's IRR (15%) is less than the WACC (16%).
- This means the project is expected to generate returns lower than the cost of financing it.
- Accepting such a project would reduce shareholder wealth and the overall value of the firm.
- The basic rule of capital budgeting is to accept projects only when  $IRR \geq WACC$  (or  $NPV \geq 0$ ).

**Question 7**

- (a) Explain why the cost of debt is always calculated on an after-tax basis, whereas the cost of preference shares is not.
- (b) A company issues 10,000 new equity shares of ₹100 each at a premium of 10%. The issue expenses (flotation cost) amount to ₹5 per share. The company pays a dividend of ₹12 per share, which is expected to grow at 5% indefinitely. Calculate the cost of new equity. If the company had used retained earnings instead of issuing new shares, how would the cost of equity change?

**Answer:**

(a)

- **Cost of Debt – After-tax Basis:**  
Interest paid on debt is a tax-deductible expense under the Income Tax Act. This means that the actual cost to the company is reduced by the tax saved. For example, if a company pays 10% interest and the tax rate is 30%, the effective after-tax cost is only 7% ( $10\% \times 0.70$ ). This tax shield makes debt cheaper than other sources of finance.
- **Cost of Preference Shares – No Tax Adjustment:**  
Dividends paid on preference shares are not considered a business expense for tax purposes. They are paid out of after-tax profits. Therefore, there is no

tax shield available, and the cost of preference shares is calculated on a pre-tax basis (i.e., the dividend rate itself).

(b)

### Step 1: Calculate Net Proceeds per Share

- Face Value = ₹100
- Premium = 10% of ₹100 = ₹10
- Issue Price = ₹110
- Flotation Cost = ₹5
- Net Proceeds (NP) = ₹110 – ₹5 = ₹105

### Step 2: Calculate Cost of New Equity

$$\begin{aligned} K_{e\_new} &= (D_1/NP) + g \\ &= 12/105 + 0.05 \\ &= 0.1143 + 0.05 \\ &= 0.1643 \text{ or } 16.43\% \end{aligned}$$

### Step 3: Cost of Retained Earnings

If retained earnings were used instead:

$$\begin{aligned} K_{e\_retained} &= D_1/P_0 + g \\ &= 12/110 + 0.05 \\ &= 0.1091 + 0.05 \\ &= 0.1591 \text{ or } 15.91\% \end{aligned}$$

### Step 4: Comparison

The cost of new equity (16.43%) is higher than the cost of retained earnings (15.91%) because flotation costs increase the effective cost of raising funds externally. Retained earnings involve no issuance costs, making them a cheaper source of equity financing.

## Question 8

**From the information provided below, compute the specific cost of each component:**

1. **Equity:** The current market price per share is ₹150. The current dividend per share is ₹13.50, and dividends are expected to grow at a rate of 6% p.a.
2. **Retained Earnings:** The shareholders' required rate of return is 15%.

3. **Debt:** The company issues 11% irredeemable debentures for ₹2,00,000 at a discount of 2%. The tax rate is 50%.
4. **Preference Shares:** The company issues 12% preference shares for ₹1,00,000 redeemable after 5 years at par. The issue expenses are 2% of the face value.

**Calculate the cost of equity, cost of retained earnings, cost of debt (after tax), and cost of preference shares.**

**Answer:**

### **1. Cost of Equity (Ke)**

$$\begin{aligned} D1 &= D0 \times (1+g) \\ &= 13.50 \times 1.06 \\ &= ₹14.31 \end{aligned}$$

$$\begin{aligned} K_e &= D1/P0 + g \\ &= 14.31/150 + 0.06 \\ &= 0.0954 + 0.06 \\ &= 0.1554 \text{ or } 15.54\% \end{aligned}$$

### **2. Cost of Retained Earnings (Kr)**

The cost of retained earnings equals the shareholders' required rate of return, as it represents the opportunity cost to existing shareholders.

$$K_r = 15\%$$

### **3. Cost of Debt (Kd) – After Tax**

- Face Value (FV) = ₹100 per debenture (assumed)
- Net Proceeds (NP) = ₹100 – 2% discount = ₹98
- Interest (I) = 11% of ₹100 = ₹11
- Tax Rate (T) = 50%

For irredeemable debt:

$$\begin{aligned} K_d &= I/NP \times (1-T) \\ &= 11/98 \times 0.50 \\ &= 0.1122 \times 0.50 \\ &= 0.0561 \text{ or } 5.61\% \end{aligned}$$

### **4. Cost of Preference Shares (Kp)**

- Face Value (FV) = ₹100 per share (assumed)
- Redemption Value (RV) = ₹100
- Net Proceeds (NP) = ₹100 – 2% issue expenses = ₹98
- Dividend (D) = 12% of ₹100 = ₹12
- Life (n) = 5 years

$$\begin{aligned}
 K_p &= [D + (RV - NP)/n] / (RV + NP) / 2 \\
 &= [12 + (100 - 98)/5] / (100 + 98) / 2 \\
 &= (12 + 0.4) / 99 \\
 &= 0.1253 \text{ or } 12.53\%
 \end{aligned}$$

### Summary of Component Costs:

| Component                 | Cost   |
|---------------------------|--------|
| Cost of Equity            | 15.54% |
| Cost of Retained Earnings | 15.00% |
| Cost of Debt (After-tax)  | 5.61%  |
| Cost of Preference Shares | 12.53% |

### Question:

A company provides the following data:

- Annual Production = 12,000 units
- Cost per unit:
  - Raw Material = ₹40
  - Labour = ₹20
  - Overheads = ₹20
- Raw material storage = 2 months
- WIP period = 1 month (50% completion for labour & overheads)
- Finished goods storage = 1 month
- Debtors collection period = 2 months
- Creditors period = 1 month
- Cash balance required = ₹50,000

Find **Working Capital Requirement**.

### Monthly Production

Annual production = 12,000 units

👉 Monthly production =  $12,000 / 12 = 1,000$  units

### Cost per Unit

- Raw Material = ₹40
  - Labour = ₹20
  - Overheads = ₹20
- 👉 Total Cost = **₹80 per unit**

### Working Capital Components

#### (A) Raw Material Stock (2 months)

$$\begin{aligned} &= 1,000 \times 2 \times 40 \\ &= \mathbf{₹80,000} \end{aligned}$$

**(B) Work-in-Progress (1 month)**

- Raw Material = 100%
- Labour & Overheads = 50%

$$\text{RM} = 1,000 \times 40 = 40,000$$

$$\begin{aligned}\text{Labour} + \text{OH} &= (20 + 20) \times 50\% = 40 \times 50\% = 20 \text{ per unit} \\ &= 1,000 \times 20 = 20,000\end{aligned}$$

👉  $\text{Total WIP} = 40,000 + 20,000 = \text{₹}60,000$

**(C) Finished Goods (1 month)**

$$\begin{aligned}&= 1,000 \times 80 \\ &= \text{₹}80,000\end{aligned}$$

**(D) Debtors (2 months)**

$$\begin{aligned}&= 1,000 \times 2 \times 80 \\ &= \text{₹}1,60,000\end{aligned}$$

**(E) Total Current Assets**

$$\begin{aligned}&= 80,000 + 60,000 + 80,000 + 1,60,000 \\ &= \text{₹}3,80,000\end{aligned}$$

**(F) Less: Creditors (1 month of RM)**

$$\begin{aligned}&= 1,000 \times 40 \\ &= \text{₹}40,000\end{aligned}$$

**(G) Net Working Capital**

$$\begin{aligned}&= 3,80,000 - 40,000 \\ &= \text{₹}3,40,000\end{aligned}$$

**(H) Add: Cash Balance**

$$= \text{₹}50,000$$



## Final Working Capital Requirement

👉 ₹3,90,000

Reliable Ltd. has given the following information: (1) Current Liability Rs. 6 Lakh (2) Fixed Assets Rs. 60 Lakh (3) Inventory Rs. 3 Lakh (4) Current Assets Rs. 15 Lakh (5) Long Term Loan Rs. 60 Lakh You are required to calculate (a) Quick Ratio, (2) Current Ratio, (3) Working Capital

#### Given Data

- Current Liabilities = ₹6 lakh
- Current Assets = ₹15 lakh
- Inventory = ₹3 lakh

#### 1. Current Ratio

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}} = \frac{15}{6} = 2.5$$

👉 Current Ratio = 2.5 : 1

#### 2. Quick Ratio (Acid-Test Ratio)

$$\text{Quick Assets} = \text{Current Assets} - \text{Inventory} = 15 - 3 = 12 \text{ lakh}$$

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}} = \frac{12}{6} = 2$$

👉 Quick Ratio = 2 : 1

#### 3. Working Capital

$$\begin{aligned} \text{Working Capital} &= \text{Current Assets} - \text{Current Liabilities} \\ &= 15 - 6 = 9 \text{ lakh} \end{aligned}$$

✅ Working Capital = ₹9 lakh

#### Final Answers

- Current Ratio = 2.5 : 1
- Quick Ratio = 2 : 1
- Working Capital = ₹9 lakh

If a company's cost of equity is 16% and the cost of debt (after tax) is 8%, with a debt-to-equity ratio of 1:1, what is the approximate WACC? (Assume no preference shares)

**Given:**

- Cost of Equity  $K_e = 16\%$
- Cost of Debt  $K_d = 8\%$  (after tax)
- Debt : Equity = 1 : 1

**Step 1: Determine Weights**

Since the ratio is 1 : 1, total = 2 parts

- Weight of Debt  $W_d = \frac{1}{2} = 0.5$
- Weight of Equity  $W_e = \frac{1}{2} = 0.5$

**Step 2: WACC Formula**

$$\text{WACC} = (W_e \times K_e) + (W_d \times K_d)$$

**Step 3: Substitute Values**

$$\begin{aligned} &= (0.5 \times 16\%) + (0.5 \times 8\%) \\ &= 8\% + 4\% = 12\% \end{aligned}$$

✅ Final Answer: WACC = 12%

**Gordon's Dividend Discount Model.**

**Gordon's Dividend Discount Model (Gordon Growth Model)** is a method used to **value equity shares** based on the present value of future dividends that grow at a constant rate.

**Formula**

$$P_0 = \frac{D_1}{K_e - g}$$

Where:

- $P_0$  = Current market price of the share
- $D_1$  = Expected dividend next year
- $K_e$  = Cost of equity (required rate of return)
- $g$  = Constant growth rate of dividends

A company's equity share is currently traded at ₹120 per share. The company is expected to pay a dividend of ₹6 per share next year, and dividends are expected to grow at a constant rate of 5% per annum. Calculate the cost of equity using the Gordon's Dividend Discount Model.

### Using Gordon's Dividend Discount Model

$$K_e = \frac{D_1}{P_0} + g$$

Given:

- $D_1 = ₹6$
- $P_0 = ₹120$
- $g = 5\% = 0.05$

Result:

$$K_e = 0.10 = 10\%$$

✓ Final Answer: Cost of Equity = 10%

### Quick Insight

Even though the dividend yield is only 5% ( $6/120$ ), adding the **growth rate (5%)** gives the total return expected by investors → 10%.



A company's equity shares are currently traded at ₹180 per share. The company expects to pay a dividend of ₹9 per share next year, and the dividends are expected to grow at a constant rate of 8% per annum.

- Calculate the cost of equity using the Gordon's Dividend Discount Model.
- If the Risk-Free Rate of return is 7%, the market rate of return is 15%, and the company's beta ( $\beta$ ) is 1.2, calculate the cost of equity using the Capital Asset Pricing Model (CAPM).
- As a financial analyst, which of the two computed costs would you consider more reliable and why?

The risk-free rate of return is 6%, the expected market return is 14%, and a company's beta ( $\beta$ ) is 1.3. Calculate the cost of equity using the Capital Asset Pricing Model (CAPM). Briefly explain what beta represents in this context.

### Using CAPM Formula

$$K_e = R_f + \beta(R_m - R_f)$$

#### Given:

- Risk-free rate  $R_f = 6\%$
- Market return  $R_m = 14\%$
- Beta  $\beta = 1.3$

✓ Final Answer: Cost of Equity = 16.4%

### What does Beta ( $\beta$ ) represent?

- Beta measures the risk (volatility) of a company's stock compared to the overall market.

#### Interpretation:

- $\beta = 1 \rightarrow$  Same risk as the market
- $\beta > 1 \rightarrow$  **More risky than market** (more volatile)
- $\beta < 1 \rightarrow$  Less risky than market

👉 In this case,  $\beta = 1.3$  means:

- The company is **30% more volatile than the market**
- Investors expect **higher returns (16.4%)** to compensate for this higher risk

#### Quick Insight

Higher beta  $\rightarrow$  higher risk  $\rightarrow$  higher required return  $\rightarrow$  higher cost of equity.

A company issues 10% preference shares of ₹100 each at a discount of 5%. The shares are redeemable after 8 years at par. Calculate the cost of preference shares using the approximation method. (Ignore flotation costs)

**Formula:**

$$K_p = \frac{D + \frac{(RV-NP)}{n}}{\frac{RV+NP}{2}}$$

**Given:**

- Face Value (RV) = ₹100
- Dividend = 10% of 100 = ₹10
- Net Proceeds (NP) = 100 – 5% = ₹95
- Redemption Value = ₹100
- Time (n) = 8 years

**Substitution:**

$$K_p = \frac{10 + \frac{(100-95)}{8}}{\frac{100+95}{2}}$$

**Final Answer:**

$$K_p \approx 10.9\%$$

**From the information provided below, compute the specific cost of each component:**

1. **Equity:** The current market price per share is ₹150. The current dividend per share is ₹13.50, and dividends are expected to grow at a rate of 6% p.a.
2. **Retained Earnings:** The shareholders' required rate of return is 15%.
3. **Debt:** The company issues 11% irredeemable debentures for ₹2,00,000 at a discount of 2%. The tax rate is 50%.
4. **Preference Shares:** The company issues 12% preference shares for ₹1,00,000 redeemable after 5 years at par. The issue expenses are 2% of the face value.

**Calculate the cost of equity, cost of retained earnings, cost of debt (after tax), and cost of preference shares.**

## 1. Cost of Equity (Ke)

Using Gordon's Dividend Discount Model:

$$K_e = \frac{D_1}{P_0} + g$$

- $D_1 = 13.50 \times (1 + 0.06) = 14.31$
- $P_0 = 150$
- $g = 6\%$

$$K_e = \frac{14.31}{150} + 0.06 = 0.0954 + 0.06 = 0.1554$$

✓ Cost of Equity = 15.54%

## 2. Cost of Retained Earnings (Kr)

Given directly:

$$K_r = 15\%$$

✓ Cost of Retained Earnings = 15%

## 3. Cost of Debt (Kd)

Irredeemable debt formula:

$$K_d = \frac{I(1 - t)}{NP}$$

- Interest = 11% of 2,00,000 = 22,000
- Tax rate = 50% → After-tax interest = 22,000 × 0.5 = 11,000
- Net Proceeds = 2,00,000 – 2% = 1,96,000

$$K_d = \frac{11,000}{1,96,000} = 0.0561$$

✓ Cost of Debt (after tax) = 5.61%

## 4. Cost of Preference Shares (Kp)

Approximation method:

$$K_p = \frac{D + \frac{(RV - NP)}{n}}{\frac{RV + NP}{2}}$$

- Dividend = 12% of 1,00,000 = 12,000
- Issue expenses = 2% → NP = 98,000
- Redemption Value (RV) = 1,00,000
- Time (n) = 5 years

$$\begin{aligned} K_p &= \frac{12,000 + \frac{(1,00,000 - 98,000)}{5}}{\frac{1,00,000 + 98,000}{2}} \\ &= \frac{12,000 + 400}{99,000} = \frac{12,400}{99,000} = 0.1253 \end{aligned}$$

✓ Cost of Preference Shares ≈ 12.53%

### ✓ Final Answers (Summary)

- Cost of Equity (Ke) = 15.54%
- Cost of Retained Earnings (Kr) = 15%
- Cost of Debt (Kd) = 5.61%
- Cost of Preference Shares (Kp) = 12.53%



A manufacturing firm has issued the following securities. Compute the cost of each component, assuming a corporate tax rate of 35%.

- Debtures: 1,000, 12% debtures of ₹100 each, redeemable after 5 years at a premium of 5%. The debtures are currently sold at ₹96 each. (Use approximation method)
- Preference Shares: 500, 10% preference shares of ₹100 each, redeemable after 4 years at par. The current market price of the preference share is ₹92. (Use approximation method)